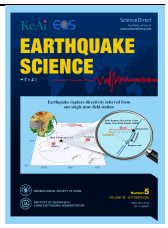
Earthquake Science 38 (2025) 450–460

Contents lists available at ScienceDirect

Earthquake Science

Journal homepage: https://www.keaipublishing.com/en/journals/earthquake-science/



Data Product

Basic processing of the InSight seismic data from Mars for further seismological research

Shuguang Wang[1,✉], Shuoxian Ning[1], Zhixiang Yao[1], Jiaqi Li[2], Wanbo Xiao[2], Tianfan Yan[3] and Feng Xu[1]

[1] *Earthquake Science Data Center*, *Institute of Geophysics*, *China Earthquake Administration*, *Beijing* 100081, *China*
[2] *School of Earth and Space Science*, *Peking University*, *Beijing* 100871, *China*
[3] *China Key Laboratory of Petroleum Resources Research*, *Institute of Geology and Geophysics*, *Chinese Academy of Sciences*, *Beijing* 100049, *China*

Key points:

- The InSight mission provided unprecedented seismic data from Mars, but with potential non-seismic noise.
- This paper presents primary processed datasets for InSight's Very Broad Band seismic data, focusing on the removal of non-seismic noise.
- The processed datasets, along with the accompanying code base, will contribute to further researches on Mars' geological properties.

A B S T R A C T

The InSight mission has obtained seismic data from Mars, offering new insights into the planet's internal structure and seismic activity. However, the raw data released to the public contain various sources of noise, such as ticks and glitches, which hamper further seismological studies. This paper presents step-by-step processing of InSight's Very Broad Band seismic data, focusing on the suppression and removal of non-seismic noise. The processing stages include tick noise removal, glitch signal suppression, multicomponent synchronization, instrument response correction, and rotation of orthogonal components. The processed datasets and associated codes are openly accessible and will support ongoing efforts to explore the geophysical properties of Mars and contribute to the broader field of planetary seismology.

Keywords: Mars; Insight; seismology; data process; seismic noise

Citation: Wang SG, Ning SX, Yao ZX, Li JQ, Xiao WB, Yan TF and Xu F (2025). Basic processing of the InSight seismic data from Mars for further seismological research. Earthq Sci **38**(5): 450–460, doi: 10.1016/j.eqs.2025.06.006.

Production and Hosting by Elsevier on behalf of KeAI

© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by/4.0).

Peer review under the responsibility of Institute of Geophysics, China Earthquake Administration.

✉ **Corresponding author.** Wang SG, email: esdc@cea-igp.ac.cn

Article history:
Received 30 August 2024
Received in revised form 21 January 2025
Accepted 23 January 2025
Available online 1 April 2025

https://doi.org/10.1016/j.eqs.2025.06.006

1. Introduction to planetary seismology and the InSight mission

China is advancing its deep-space exploration with a particular focus on lunar and Martian missions (Pan WX and Wang C, 2021; Wan WX et al., 2019; Zhang RQ et al., 2022). Planetary science is an interdisciplinary field that investigates the fundamental characteristics, formation, and evolutionary processes of planets and their moons. Among various techniques used in this field, planetary seismology is one of the most direct methods for probing the internal structures of solid celestial bodies such as the Moon and Mars. Much like international missions such as India's Chandrayaan-3 (which landed on the Moon in 2023), the United States' Farside Seismic Suite and Artemis missions to the Moon, and the Dragonfly mission to Titan, China's Chang'e 7 mission is equipped with seismological instruments designed to explore the internal structures of these celestial bodies.

Mars, one of the most Earth-like planets in the Solar System, has significant implications for understanding the evolution of terrestrial planets and searching for extraterrestrial life. Since Mariner 4's historic flyby of Mars in 1965, which provided the first images of the Martian surface, the exploration of Mars has continued for nearly six decades, expanding our knowledge of Martian topography, gravity, magnetic fields, and rock composition.

In 1976, Viking 2 became the first successful Mars lander equipped with a seismometer. However, the seismometer was mounted on the lander, which made it highly susceptible to wind-induced vibrations, and it only recorded one signal that was regarded as a possible seismic event (Anderson et al., 1977; Lazarewicz, 2023). It was not until the successful landing of the InSight mission on Mars in 2018 that high-quality seismic data became available, enabling investigations of the Martian interior through seismological methods (Banerdt et al., 2020; Giardini et al., 2020; Golombek et al., 2020; Lognonné et al., 2020).

The InSight mission, short for " Interior Exploration using Seismic Investigations, Geodesy and Heat Transport", aims to explore Mars' interior structure via seismic waves, geodesy, and heat flow. The mission's primary scientific goals are to obtain data on the internal structure of Mars to infer its evolutionary history and determine the current levels of tectonic activities and impact cratering (Banerdt et al., 2020). InSight is equipped with a suite of scientific instruments for this purpose, including the Seismic Experiment for Internal Structure (SEIS), Heat Flow and Physical Properties Package (HP^3), Rotation and Interior Structure Experiment (RISE), and an Auxiliary Payload Sensor Suite (APSS) that includes the Pressure Sensor (PS) and Temperature and Winds for InSight (TWINS) instruments, among others.

The SEIS instrument comprises four main components (Lognonné et al., 2019): the sensor assembly, wind and thermal shield, tether, and electronic box. The sensor assembly included a very broadband (VBB) three-component seismometer and a short-period three-component seismometer.

On November 26, 2018, InSight landed on Mars near the equator in the Elysium Planitia region (4.5° N, 135.6° E) (Figure 1). The sensor assembly was deployed directly on the surface to the south of the lander, using a robotic arm. The SEIS instrument is equipped with a leveling system that can accommodate a ground slope of up to 15°. The wind and thermal shield was positioned above the sensor to reduce interference from variations in temperature and atmospheric activity on Mars. The SEIS was connected via the tether to the lander's internal electronic box for signal transmission and processing.

The SEIS seismometer of the InSight mission ceased data collection on December 22, 2022, owing to a power shortage after operating for 1,447 Martian sols (a Martian day is 24 h, 39 min, and 35 s). Owing to limitations in data transmission between Mars and Earth, only downsampled data were continuously transmitted at 20 samples per second (sps), whereas high-sampling-rate data (100 sps) were only transmitted during significant marsquake events. Marsquake events are currently classified based on the frequency range of the seismic wave energy and the signal-to-noise ratio. Most marsquakes exhibit narrow frequency bands and low signal-to-noise ratios, making it challenging to identify P- and S-waves clearly. Among more than 1,300 marsquake or impact events cataloged by the MarsQuake Service (MQS; InSight Marsquake Service, 2023), only a few high-quality events have been accurately identified (Ceylan et al., 2023; Wang X et al., 2023).

Seismological data from the InSight mission are crucial for advancing our understanding of Mars' seismic activity, shallow structure, and the composition of its crust, mantle, and core (Liu LB, 2021; Sun WJ et al., 2021; Lognonné et al., 2023). Although all seismological data obtained by InSight have been made publicly available, processing and using the data is challenging, especially the suppression of glitch noise and the removal of tick noise (Compaire et al., 2021; Scholz et al., 2020; Xu WC et al., 2022). This study offers a step-by-step approach for processing InSight seismological data and provides a directly accessible shared database to support further research by scientists in the field.

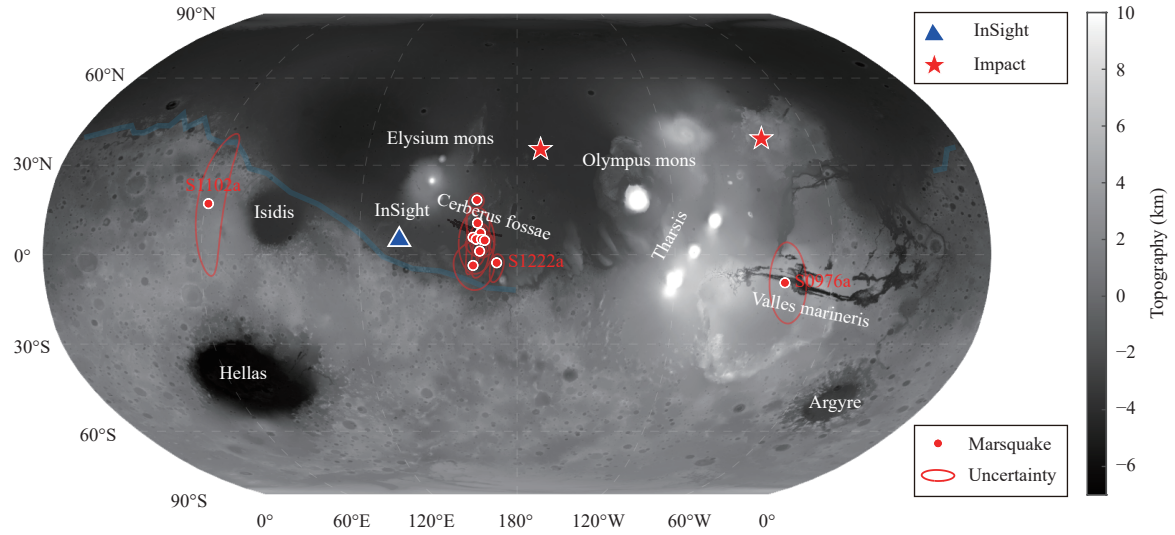


Figure 1. Global topographic map of Mars (Smith et al., 2003). The blue triangle marks the landing site of the InSight mission, the red circles indicate detected marsquakes (with ellipses representing the uncertainty in their azimuthal angle), and the red stars denote meteoroid impact events identified from the seismometer data and remote sensing images.

2. InSight seismic data

Raw seismic data from the InSight mission, specifically from the 0.01−50 Hz VBB sensors, are publicly accessible via IRIS (https://www.iris.edu/hq/sis/insight), NASA PDS (https://pds-geosciences.wustl.edu/missions/insight/seis.htm) and IPGP (InSight Mars SEIS Data Service, 2019a, b). The VBB sensors (0.01−50 Hz) are more suitable for scientific analyses than the short-period sensors (0.1−50 Hz) as they offer a broader frequency coverage, higher time resolution, and lower self-noise levels below 5 Hz. Therefore, this study concentrated on processing VBB data within the 0.01−50 Hz range.

Noise in the VBB data can impact seismological analyses in several ways (Ceylan et al., 2021, 2022) and is generally categorized into two types. The first type includes instrument self-noise, environmental noise caused by lander modes, and transient high-frequency noise (i.e., donk), which were not eliminated during the data processing. The second category includes glitches and tick noise, which were addressed in this study (Figure 2). Additionally, the three-component records (U, V, and W) tilted relative to the horizontal plane and not mutually orthogonal must be rotated into the *Z*-N- E coordinate system for VBB data processing.

A self-noise model, previously established for different components of the VBB instrument (Lognonné et al., 2020), was compared with the observed seismic data to identify non-self-noise signals in the seismic database. The results showed that environmental noise was significantly higher than the instrument's self-noise level during the Martian daytime, but quieter during the nighttime when it approached the instrument's self-noise level (Figure 2). MQS, the official marsquake catalog publisher for InSight, states that detecting marsquakes during the daytime is challenging owing to wind noise, with 99% of recorded marsquakes occurring at night (Clinton et al., 2021). The occurrence of marsquakes during the Martian daytime remains unknown and requires further investigation.

Lander mode vibrations caused by the wind are a prominent environmental noise source during the daytime and are characterized by discrete narrow energy bands such as 3.2, 4.0, 7.0, and 11.7 Hz (Figure 2). These vibrations exhibit strong energy levels and frequencies that correlate with the local air temperature, impacting seismological analyses such as the H/V spectral ratio and time-frequency analysis (Xiao WB and Wang YB, 2021; Dahmen et al., 2021). Although these vibrations have high frequencies that minimally affect marsquake analyses, they can still be used to estimate the impact of the air temperature on the elastic modulus.

Transient high-frequency signals (donk) typically appear above 12 Hz and resemble microseisms, but last for only a few seconds. These events may be related to instrument stress release due to temperature cycles (Ceylan et al., 2021), and each one must be identified based on their waveform and spectral characteristics to avoid misinterpreting the seismic data. Donk mainly affects high-frequency components and has minimal overlap with the marsquake signal frequency range, thus affecting the short-period sensors more significantly. Consequently, we focused on glitch noise in this study because it has low-frequency components that may affect the marsquake recordings.

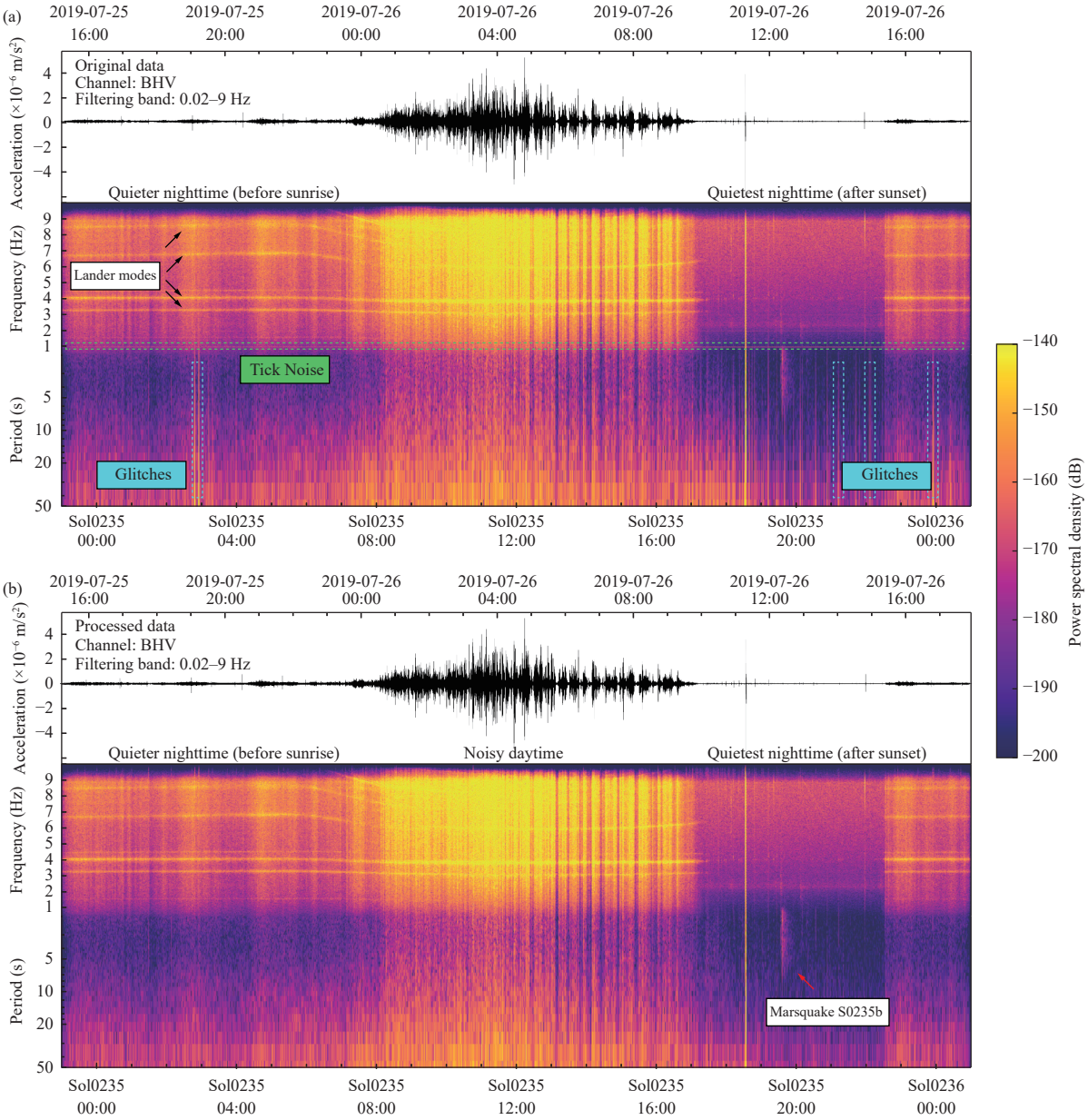


Figure 2. Waveforms and time-frequency plots of InSight's VBB seismic data recorded on Sol 0235. (a) shows the raw data, while (b) shows the processed data after suppression of glitch noise and removal of tick noise.

Glitches are primarily associated with thermal stress release owing to temperature variations and feature low-frequency components that appear as transient single-sided pulse signals with potentially large amplitudes. Glitches can affect an extensive range of seismological methods, including phase picking, receiver functions, polarity analysis, and autocorrelation analysis. Established methods for suppressing glitches are available (Scholz et al., 2020; Xu WC et al., 2022), with both glitch-affected and glitch-suppressed VBB data, as illustrated in Figures 2 and 3.

Tick noise, caused by internal electronic crosstalk within the seismometer, manifests at 1 Hz and its higher harmonics (e.g., 2 Hz and 4 Hz). This type of noise interferes significantly with signals near 1 Hz. If not properly addressed, it can adversely affect seismological studies, such as marsquake spectral analysis and estimation of source-corner frequencies (Figure 4). Therefore, effective removal of this noise is crucial for ensuring the accuracy and reliability of these analyses.

3. Data processing method

We downloaded miniSEED-format data at a rate of 20 sps from publicly available databases and performed a series of processing steps, including the removal of tick

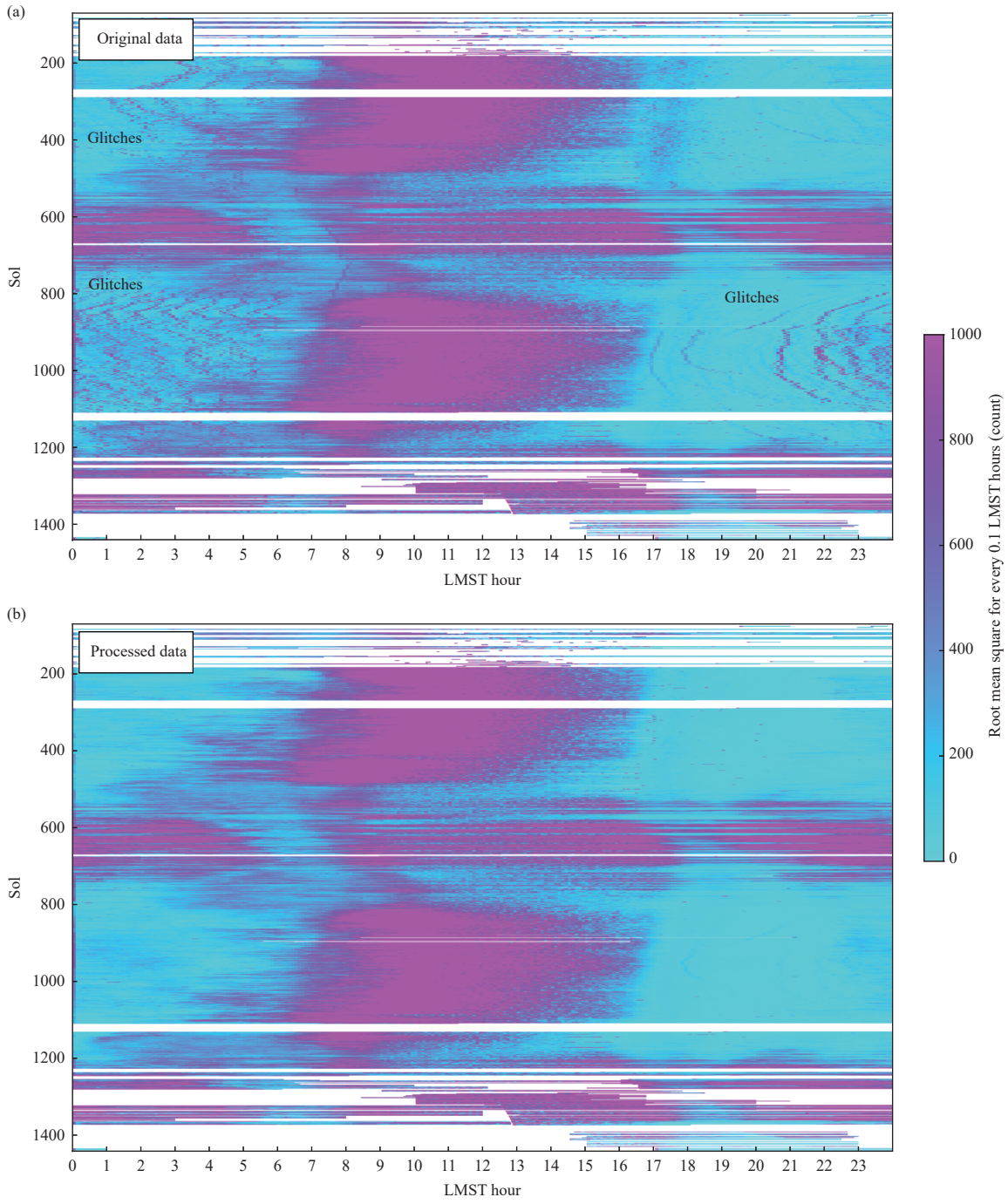


Figure 3. Maximum amplitude distribution of VBB seismic data during the operations of the InSight mission. (a) shows the raw data, while (b) shows the processed data after suppression of glitch noise and removal of tick noise, LMST: Local Mean Solar Time.

noise, glitch signal suppression, multi-component synchronization, instrument response correction, and orthogonal rotation (Figure 5). The downloaded data were segmented according to the Martian day, with each segment covering one hour before and one hour after the sol (e.g., data for Sol 235 were segmented from Sol 234, 23:00 LMST to Sol 236, 01:00 LMST).

Step 1 - Removal of tick noise: Tick noise originates from electronic signal interference within the SEIS seismometer, with consistent patterns and amplitudes throughout the mission. A method for removing tick noise was implemented in this study based on the method proposed by Compaire et al. (2021). We first identified the tick noise for the U, V, and W components by stacking continuous waveforms (the stacked tick-noise is stored in '. /Utilities/tk_noise/' as an .npy format file). Next, we removed it

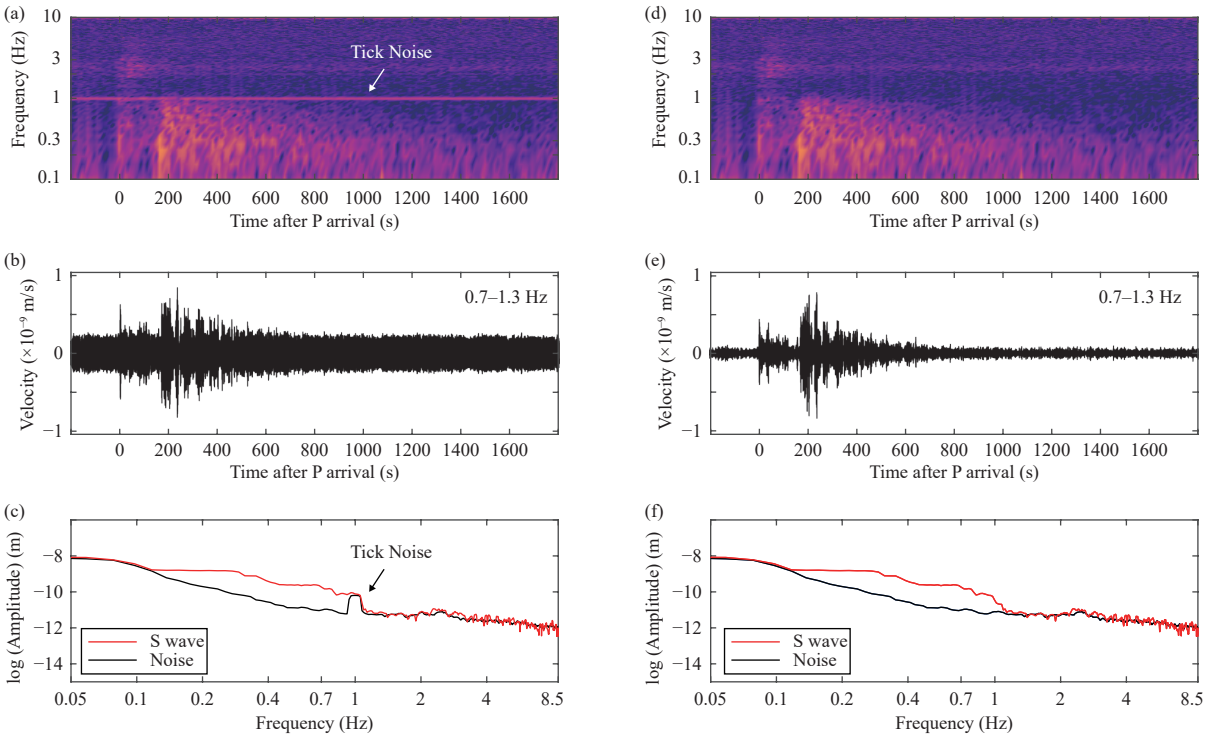


Figure 4. Comparison of the data before and after tick-noise removal. (a) Time-frequency spectrum, (b) time-domain waveform, and (c) S-wave spectrum of the S0235b event. The left side (a−c) shows the raw data, while the right side (d−f) presents the data after the removal of the tick noise.

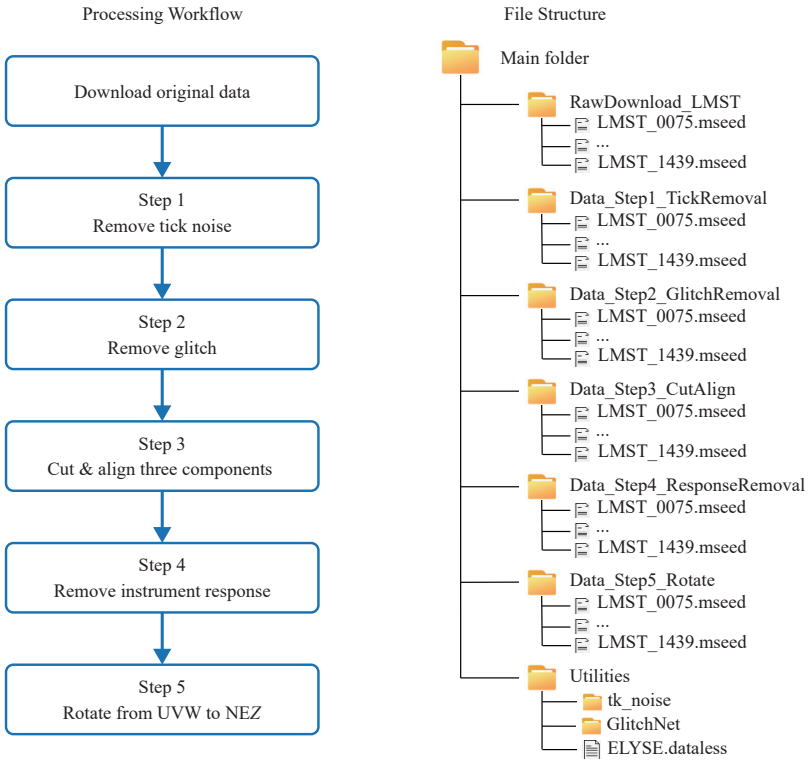


Figure 5. Data processing workflow and the file structure of the available data.

from the continuous waveforms by subtracting the stacked tick-noise (using the 'remove_tick_noise' function).

Step 2 - Suppression of glitch signals: Glitch signals were suppressed using the deep learning-based GlitchNet

neural network (Xu WC et al., 2022).

Step 3 - Multi-component synchronization: Because the U, V, and W components may have partial data gaps, we selected the data segments where all three components were recorded and used these for trimming and time alignment to ensure that the synchronized data spanned the same time period.

Step 4 - Instrument response correction: Instrument response was corrected to convert the data from the raw count unit to velocity records for the U, V, and W components. Pre-filtering of 0.01−9.0 Hz was applied to enhance stability during the deconvolution.

Step 5 - Orthogonal rotation: Using ObsPy (Beyreuther et al., 2010), the non-orthogonal U, V, and W components were rotated to the orthogonal N, E, and *Z* components.

4. Results: Seismic noise on Mars, Earth, and the Moon

The noise level of the InSight seismic data is revealed by the power spectral densities (PSD) of the vertical component during a typical sol (Figure 6) for the period range 0.1−100 s (0.01−10 Hz). The PSD levels vary with local time, exhibiting higher levels in the local daytime and lower levels in the local nighttime. In general, the PSD amplitudes are significantly higher for periods below approximately 2.5 s (above 0.4 Hz). During the noisy daytime, the noise level of short-period features (below 10 s) is 1−3 or ders of magnitude higher on Mars than that on the Moon. Some sharp peaks were visible at short periods, which were attributed to contaminations by mechanical noise such as wind-induced lander vibrations (Dahmen et al., 2021). The vertical gap at 1.0 s in the PSD is due to the removal of the tick noise. During the quiet nighttime, the noise level is much lower than that in the daytime and is also lower than the Earth Low Noise Model (Peterson, 1993) at 0.3−10 s (0.1−3.3 Hz). The lowest nighttime PSD has an acceleration of 1.5×10^{-10} m·s^2·Hz$^{-1/2}$ at 2.5 s (0.4 Hz), about 1/500 of the Earth Low Noise Model (Figure 6; Lognonné et al., 2020). The short-period part of the nighttime records is very similar to the VBB self-noise level, which indicates that during the nighttime, the VBB sensor mainly records its self-noise as short-period modes. The acute variation of the noise level on Mars at different local times requires additional caution when interpreting seismic data from the InSight mission.

5. Outlook for Martian seismic data

Although seismicity on Mars is much lower than on Earth, possibly indicating a lack of plate tectonics, the number of tectonic seismic events on Mars is notably higher than that of shallow tectonic moonquakes (Banerdt et al., 2020). Over its four Martian years of operation, SEIS recorded more than 1,300 seismic events (Ceylan et al., 2022), offering a wealth of valuable data for studying the internal structure and composition of Mars. These data provide insight into the shallow subsurface (Hobiger, 2021), crust (Knapmeyer-Endrun et al., 2021; Li JQ et al., 2022a, b), mantle (Khan et al., 2021), and core (Stähler et al., 2021; Khan et al., 2023; Samuel et al., 2023) of Mars.

Additionally, integrated analyses of seismic data with atmospheric parameters, such as pressure and wind speed, have deepened our understanding of the physical processes in Mars' lower atmosphere (Banfield et al., 2020). These discoveries not only enhance our knowledge of Mars' internal structure and climate system, but also provide crucial scientific support for assessing the feasibility of Mars as a potentially habitable planet.

6. Summary

This paper provides an overview of seismic data from the Insight mission and details the step-by-step processing

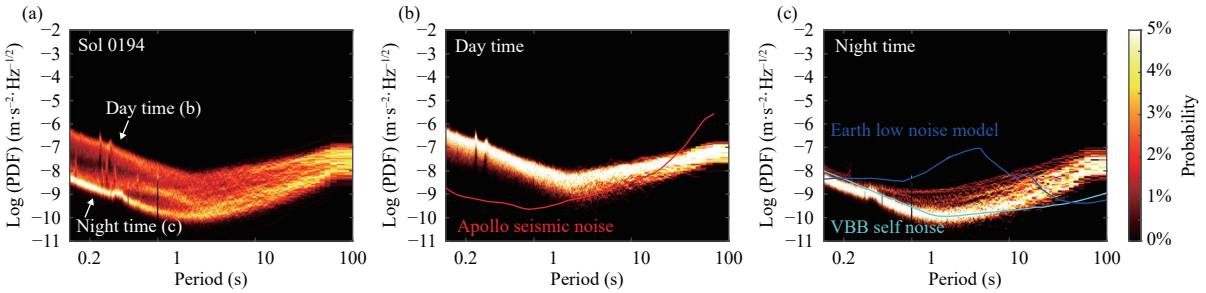


Figure 6. Statistical power spectral density levels of InSight seismic data for the vertical component during (a) the daytime and nighttime, (b) only daytime, and (c) only nighttime on Sol 194 . The Apollo seismic noise (in red) and VBB self-noise (in cyan) are from Lognonné et al. (2019, 2020). The Earth low noise model (in blue) is from Peterson (1993). Note that the horizontal axis shows the period (reciprocal to frequency).

of VBB seismic data. The processing workflow includes suppressing glitch noise, removing tick noise, synchronizing the three seismic components, and rotating them to Z-N-E components.

The resulting dataset is more accessible to seismologists who may be less experienced with Martian seismic data. These processed data not only facilitate advanced seismic analyses, but also support integration with other Martian environmental datasets. Moreover, it can be compared with seismic data from other planets, including Earth, advancing our understanding of planetary geophysics.

7. Data download instructions

The data, stored in miniSEED format and organized by Martian day, include both raw and processed datasets for each processing step. The processing stages include tick noise removal, glitch signal suppression, multicomponent synchronization, instrument response correction, and rotation to orthogonal components. In addition, we have provided programs for reproducing each processing step.

The dataset is available at the website (www.esdc.ac.cn/article/170) maintained by the International Earthquake Science Data Center (IEC), and can also be accessed through SciDB (https://www.scidb.cn/s/IVjuUz).

Acknowledgements

We thank Dr. Wuchuan Xu (Peking University) for assistance with deglitching and tick noise removal. We acknowledge NASA, CNES, their partner agencies and institutions (UKSA, SSO, DLR, JPL, IPGP-CNRS, ETHZ, IC, and MPS-MPG), and the flight operations teams at JPL, SISMOC, MSDS, IRIS-DMC, and PDS for providing the SEED SEIS data. This study was supported by the National Key R&D Program of China (Nos. 2022YFF0503203 and 2024YFF0809900), the Research Funds of the Institute of Geophysics, China Earthquake Administration (No. DQJB24X28), and the National Natural Science Foundation of China (Nos. 42474226 and 42441827).

Conflict of interest

The authors affirm that they have no financial and personal relationships with any individuals or organization that could have potentially influenced the work presented in this paper.

References

Anderson DL, Miller WF, Latham GV, Nakamura Y, Toksöz MN, Dainty AM, Duennebier FK, Lazarewicz AR, Kovach RL and Knight TCD (1977). Seismology on Mars. J Geophys Res Atmos **82**(28): 4524–4546. https://doi.org/10.1029/js082i028p04524.

Banerdt WB, Smrekar SE, Banfield D, Giardini D, Golombek M, Johnson CL, Lognonné P, Spiga A, Spohn T, Perrin C, Stähler SC, Antonangeli D, Asmar S, Beghein C, Bowles N, Bozdag E, Chi P, Christensen U, Clinton J, Collins GS, Daubar I, Dehant V, Drilleau M, Fillingim M, Folkner W, Garcia RF, Garvin J, Grant J, Grott M, Grygorczuk J, Hudson T, Irving JCE, Kargl G, Kawamura T, Kedar S, King S, Knapmeyer-Endrun B, Knapmeyer M, Lemmon M, Lorenz R, Maki JN, Margerin L, Mclennan SM, Michaut C, Mimoun D, Mittelholz A, Mocquet A, Morgan P, Mueller NT, Murdoch N, Nagihara S, Newman C, Nimmo F, Panning M, Pike WT, Plesa AC, Rodriguez S, Rodriguez-Manfredi JA, Russell CT, Schmerr N, Siegler M, Stanley S, Stutzmann E, Teanby N, Tromp J, Van Driel M, Warner N, Weber R and Wieczorek M (2020). Initial results from the InSight mission on Mars. Nat Geosci **13**(3): 183–189. https://doi.org/10.1038/s41561-020-0544-y.

Banfield D, Spiga A, Newman C, Forget F, Lemmon M, Lorenz R, Murdoch N, Viudez-Moreiras D, Pla-Garcia J, Garcia RF, Lognonné P, Karatekin Ö, Perrin C, Martire L, Teanby N, Hove BV, Maki JN, Kenda B, Mueller NT, Rodriguez S, Kawamura T, Mcclean JB, Stott AE, Charalambous C, Millour E, Johnson CL, Mittelholz A, Määttänen A, Lewis SR, Clinton J, Stähler SC, Ceylan S, Giardini D, Warren T, Pike WT, Daubar I, Golombek M, Rolland L, Widmer-Schnidrig R, Mimoun D, Beucler É, Jacob A, Lucas A, Baker M, Ansan V, Hurst K, Mora-Sotomayor L, Navarro S, Torres J, Lepinette A, Molina A, Marin-Jimenez M, Gomez-Elvira J, Peinado V, Rodriguez-Manfredi JA, Carcich BT, Sackett S, Russell CT, Spohn T, Smrekar SE and Banerdt WB (2020). The atmosphere of Mars as observed by InSight. Nat Geosci **13**(3): 190–198. https://doi.org/10.1038/s41561-020-0534-0.

Beyreuther M, Barsch R, Krischer L, Megies T, Behr Y and Wassermann J (2010). ObsPy: A Python toolbox for seismology. Seismol Res Lett **81**(3): 530–533. https://doi.org/10.1785/gssrl.81.3.530.

Ceylan S, Clinton JF, Giardini D, Böse M, Charalambous C, Driel MV, Horleston A, Kawamura T, Khan A, Orhand-Mainsant G, Scholz JR, Stähler SC, Euchner F, Banerdt WB, Lognonné P, Banfield D, Beucler E, Garcia RF, Kedar S, Panning MP, Pike WT, Smrekar SE, Spiga A, Dahmen NL, Hurst K, Stott AE, Lorenz RD, Schimmel M, Stutzmann E, Pierick JT, Conejero V, Pardo C and Perrin C (2021). Companion guide to the marsquake catalog from InSight, Sols 0–478: data content and non-seismic events. Phys Earth Planet Inter **310**: 106597. https://doi.org/10.1016/j.pepi.2020.106597.

Ceylan S, Clinton JF, Giardini D, Stähler SC, Horleston A,

Kawamura T, Böse M, Charalambous C, Dahmen NL, Van Driel M, Durán C, Euchner F, Khan A, Kim D, Plasman M, Scholz JR, Zenhäusern G, Beucler E, Garcia RF, Kedar S, Knapmeyer M, Lognonné P, Panning MP, Perrin C, Pike WT, Stott AE and Banerdt WB (2022). The marsquake catalogue from InSight, sols 0–1011. Phys Earth Planet Inter **333**: 106943. https://doi.org/10.1016/j.pepi.2022.106943.

Ceylan S, Giardini D, Clinton JF, Kim D, Khan A, Stähler SC, Zenhäusern G, Lognonné P and Banerdt WB (2023). Mapping the seismicity of Mars with InSight. J Geophys Res: Planets **128**(8): e2023JE007826. https://doi.org/10.1029/2023JE007826.

Clinton JF, Ceylan S, van Driel M, Giardini D, Stähler SC, Böse M, Charalambous C, Dahmen NL, Horleston A, Kawamura T, Khan A, Orhand-Mainsant G, Scholz JR, Euchner F, Banerdt WB, Lognonné P, Banfield D, Beucler E, Garcia RF, Kedar S, Panning MP, Perrin C, Pike WT, Smrekar SE, Spiga A and Stott AE (2021). The Marsquake catalogue from InSight, sols 0–478. Phys Earth Planet Inter **310**: 106595. https://doi.org/10.1016/j.pepi.2020.106595.

Compaire N, Margerin L, Garcia RF, Pinot B, Calvet M, Orhand‐mainsant G, Kim D, Lekic V, Tauzin B, Schimmel M, Stutzmann E, Knapmeyer‐Endrun B, Lognonné P, Pike WT, Schmerr N, Gizon L and Banerdt WB (2021). Autocorrelation of the ground vibrations recorded by the SEIS-InSight seismometer on Mars. J Geophys Res: Planets **126**(4): e2020JE006498. https://doi.org/10.1029/2020JE006498.

Dahmen NL, Zenhäusern G, Clinton JF, Giardini D, Stähler SC, Ceylan S, Charalambous C, Van Driel M, Hurst KJ, Kedar S, Lognonné P, Murdoch N, Myhill R, Panning MP, Pike WT, Schimmel M, Schmelzbach C, Scholz JR, Stott AE, Stutzmann E and Banerdt WB (2021). Resonances and lander modes observed by InSight on Mars (1–9 Hz). Bull Seismol Soc Am **111**(6): 2924–2950. https://doi.org/10.1785/0120210056.

Giardini D, Lognonné P, Banerdt WB, Pike WT, Christensen U, Ceylan S, Clinton JF, Van Driel M, Stähler SC, Böse M, Garcia RF, Khan A, Panning M, Perrin C, Banfield D, Beucler E, Charalambous C, Euchner F, Horleston A, Jacob A, Kawamura T, Kedar S, Mainsant G, Scholz JR, Smrekar SE, Spiga A, Agard C, Antonangeli D, Barkaoui S, Barrett E, Combes P, Conejero V, Daubar I, Drilleau M, Ferrier C, Gabsi T, Gudkova T, Hurst K, Karakostas F, King S, Knapmeyer M, Knapmeyer-Endrun B, Llorca-Cejudo R, Lucas A, Luno L, Margerin L, Mcclean JB, Mimoun D, Murdoch N, Nimmo F, Nonon M, Pardo C, Rivoldini A, Manfredi JAR, Samuel H, Schimmel M, Stott AE, Stutzmann E, Teanby N, Warren T, Weber RC, Wieczorek M and Yana C (2020). The seismicity of Mars. Nat Geosci **13**(3): 205–212. https://doi.org/10.1038/s41561-020-0539-8.

Golombek M, Warner NH, Grant JA, Hauber E, Ansan V, Weitz CM, Williams N, Charalambous C, Wilson SA, Demott A, Kopp M, Lethcoe-Wilson H, Berger L, Hausmann R, Marteau E, Vrettos C, Trussell A, Folkner W, Le Maistre S, Mueller N, Grott M, Spohn T, Piqueux S, Millour E, Forget F, Daubar I, Murdoch N, Lognonné P, Perrin C, Rodriguez S, Pike WT, Parker T, Maki J, Abarca H, Deen R, Hall J andres P, Ruoff N, Calef F, Smrekar S, Baker MM, Banks M, Spiga A, Banfield D, Garvin J, Newman CE and Banerdt WB (2020). Geology of the InSight landing site on Mars. Nat Commun **11**(1): 1014. https://doi.org/10.1038/s41467-020-14679-1.

Hobiger M, Hallo M, Schmelzbach C, Stähler SC, Fäh D, Giardini D, Golombek M, Clinton J, Dahmen N, Zenhäusern G, Knapmeyer-Endrun B, Carrasco S, Charalambous C, Hurst K, Kedar S and Banerdt WB (2021). The shallow structure of Mars at the InSight landing site from inversion of ambient vibrations. Nat Commun **12**(1): 6756. https://doi.org/10.1038/s41467-021-26957-7.

InSight Mars SEIS Data Service (2019a). Data service, InSight SEIS data Bundle. PDS Geosciences (GEO) Node. Accessed August 30th, 2024. https://doi.org/10.17189/1517570.

InSight Mars SEIS Data Service (2019b). SEIS raw data, Insight Mission. IPGP, JPL, CNES, ETHZ, ICL, MPS, ISAE-Supaero, LPG, MFSC. Accessed August 30th, 2024. https://doi.org/10.18715/SEIS.INSIGHT.XB_2016.

InSight Marsquake Service (2023). Mars seismic catalogue, InSight mission; V14 2023/04/01. ETHZ, IPGP, JPL, ICL, ISAE-Supaero, MPS, Univ. Bristol. Accessed August 3th, 2024. https://doi.org/10.12686/a21.

Khan A, Ceylan S, van Driel M, Giardini D, Lognonné P, Samuel H, Schmerr NC, Stähler SC, Duran AC, Huang QC, Kim D, Broquet A, Charalambous C, Clinton JF, Davis PM, Drilleau M, Karakostas F, Lekic V, Mclennan SM, Maguire RR, Michaut C, Panning MP, Pike WT, Pinot B, Plasman M, Scholz JR, Widmer-Schnidrig R, Spohn T, Smrekar SE and Banerdt WB (2021). Upper mantle structure of Mars from InSight seismic data. Science **373**(6553): 434–438. https://doi.org/10.1126/science.abf2966.

Khan A, Huang D, Durán C, Sossi PA, Giardini D and Murakami M (2023). Evidence for a liquid silicate layer atop the Martian core. Nature **622**(7984): 718–723. https://doi.org/10.1038/s41586-023-06586-4.

Knapmeyer-Endrun B, Panning MP, Bissig F, Joshi R, Khan A, Kim D, Lekić V, Tauzin B, Tharimena S, Plasman M, Compaire N, Garcia RF, Margerin L, Schimmel M, Stutzmann É, Schmerr N, Bozdağ E, Plesa AC, Wieczorek MA, Broquet A, Antonangeli D, Mclennan SM, Samuel H, Michaut C, Pan L, Smrekar SE, Johnson CL, Brinkman N, Mittelholz A, Rivoldini A, Davis PM, Lognonné P, Pinot B, Scholz JR, Stähler S, Knapmeyer M, Van Driel M, Giardini D and Banerdt WB (2021). Thickness and structure of the Martian crust from InSight seismic data. Science **373**(6553): 438–443. https://doi.org/10.1126/science.abf8966.

Lazarewicz AR (2023). Viking marsquakes 1976 —seismic archaeology. J Geophys Res: Planets **128**(7): e2022JE007660. https://doi.org/10.1029/2022JE007660.

Li JQ, Beghein C, Wookey J, Davis P, Lognonné P, Schimmel M, Stutzmann E, Golombek M, Montagner JP and Banerdt

WB (2022a). Evidence for crustal seismic anisotropy at the InSight lander site. Earth Planet Sci Lett **593**: 117654. https://doi.org/10.1016/j.epsl.2022.117654.

Li JQ, Beghein C, McLennan SM, Horleston AC, Charalambous C, Huang QC, Zenhäusern G, Bozdağ E, Pike WT, Golombek M, Lekić V, Lognonné P and Bruce Banerdt W (2022b). Constraints on the Martian crust away from the InSight landing site. Nat Commun **13**(1): 7950. https://doi.org/10.1038/s41467-022-35662-y.

Liu LB (2021). Will InSight be able to accomplish the task to image Mars interior? Rev Geophys Planet Phys 52(1): 115–122. https://doi.org/10.19975/j.dqyxx.2020-015 (in Chinese with English abstract).

Lognonné P, Banerdt WB, Giardini D, Pike WT, Christensen U, Laudet P, de Raucourt S, Zweifel P, Calcutt S, Bierwirth M, Hurst KJ, Ijpelaan F, Umland JW, Llorca-Cejudo R, Larson SA, Garcia RF, Kedar S, Knapmeyer-Endrun B, Mimoun D, Mocquet A, Panning MP, Weber RC, Sylvestre-Baron A, Pont G, Verdier N, Kerjean L, Facto LJ, Gharakanian V, Feldman JE, Hoffman TL, Klein DB, Klein K, Onufer NP, Paredes-Garcia J, Petkov MP, Willis JR, Smrekar SE, Drilleau M, Gabsi T, Nebut T, Robert O, Tillier S, Moreau C, Parise M, Aveni G, Ben Charef S, Bennour Y, Camus T, Dandonneau PA, Desfoux C, Lecomte B, Pot O, Revuz P, Mance D, tenPierick J, Bowles NE, Charalambous C, Delahunty AK, Hurley J, Irshad R, Liu H, Mukherjee AG, Standley IM, Stott AE, Temple J, Warren T, Eberhardt M, Kramer A, Kühne W, Miettinen EP, Monecke M, Aicardi C andré M, Baroukh J, Borrien A, Bouisset A, Boutte P, Brethomé K, Brysbaert C, Carlier T, Deleuze M, Desmarres JM, Dilhan D, Doucet C, Faye D, Faye-Refalo N, Gonzalez R, Imbert C, Larigauderie C, Locatelli E, Luno L, Meyer JR, Mialhe F, Mouret JM, Nonon M, Pahn Y, Paillet A, Pasquier P, Perez G, Perez R, Perrin L, Pouilloux B, Rosak A, Savin de Larclause I, Sicre J, Sodki M, Toulemont N, Vella B, Yana C, Alibay F, Avalos OM, Balzer MA, Bhandari P, Blanco E, Bone BD, Bousman JC, Bruneau P, Calef FJ, Calvet RJ, D'Agostino SA, de Los Santos G, Deen RG, Denise RW, Ervin J, Ferraro NW, Gengl HE, Grinblat F, Hernandez D, Hetzel M, Johnson ME, Khachikyan L, Lin JY, Madzunkov SM, Marshall SL, Mikellides IG, Miller EA, Raff W, Singer JE, Sunday CM, Villalvazo JF, Wallace MC, Banfield D, Rodriguez-Manfredi JA, Russell CT, Trebi-Ollennu A, Maki JN, Beucler E, Böse M, Bonjour C, Berenguer JL, Ceylan S, Clinton J, Conejero V, Daubar I, Dehant V, Delage P, Euchner F, Estève I, Fayon L, Ferraioli L, Johnson CL, Gagnepain-Beyneix J, Golombek M, Khan A, Kawamura T, Kenda B, Labrot P, Murdoch N, Pardo C, Perrin C, Pou L, Sauron A, Savoie D, Stähler S, Stutzmann E, Teanby NA, Tromp J, van Driel M, Wieczorek M, Widmer-Schnidrig R, Wookey J (2019). SEIS: Insight's seismic experiment for internal structure of Mars. Space Sci Rev **215**(1): 12. https://doi.org/10.1007/s11214-018-0574-6.

Lognonné P, Banerdt WB, Pike WT, Giardini D, Christensen U, Garcia RF, Kawamura T, Kedar S, Knapmeyer-Endrun B, Margerin L, Nimmo F, Panning M, Tauzin B, Scholz JR, Antonangeli D, Barkaoui S, Beucler E, Bissig F, Brinkman N, Calvet M, Ceylan S, Charalambous C, Davis P, Van Driel M, Drilleau M, Fayon L, Joshi R, Kenda B, Khan A, Knapmeyer M, Lekic V, Mcclean J, Mimoun D, Murdoch N, Pan L, Perrin C, Pinot B, Pou L, Menina S, Rodriguez S, Schmelzbach C, Schmerr N, Sollberger D, Spiga A, Stähler S, Stott A, Stutzmann E, Tharimena S, Widmer-Schnidrig R andersson F, Ansan V, Beghein C, Böse M, Bozdag E, Clinton J, Daubar I, Delage P, Fuji N, Golombek M, Grott M, Horleston A, Hurst K, Irving J, Jacob A, Knollenberg J, Krasner S, Krause C, Lorenz R, Michaut C, Myhill R, Nissen-Meyer T, Ten Pierick J, Plesa AC, Quantin-Nataf C, Robertsson J, Rochas L, Schimmel M, Smrekar S, Spohn T, Teanby N, Tromp J, Vallade J, Verdier N, Vrettos C, Weber R, Banfield D, Barrett E, Bierwirth M, Calcutt S, Compaire N, Johnson CL, Mance D, Euchner F, Kerjean L, Mainsant G, Mocquet A, Rodriguez Manfredi JA, Pont G, Laudet P, Nebut T, De Raucourt S, Robert O, Russell CT, Sylvestre-Baron A, Tillier S, Warren T, Wieczorek M, Yana C and Zweifel P (2020). Constraints on the shallow elastic and anelastic structure of Mars from InSight seismic data. Nat Geosci **13**(3): 213–220. https://doi.org/10.1038/s41561-020-0536-y.

Lognonné P, Banerdt WB, Clinton J, Garcia RF, Giardini D, Knapmeyer-Endrun B, Panning M and Pike WT (2023). Mars seismology. Ann Rev Earth Planet Sci **51**(1): 643–670. https://doi.org/10.1146/annurev-earth-031621-073318.

Pan YX and Wang C (2021). Developing the planetary science research for the sustainable deep space exploration of China. Bull Natl Nat Sci Found China **35**(2): 181–185. https://doi.org/10.16262/j.cnki.1000-8217.2021.02.004 (in Chinese with English abstract).

Peterson JR (1993). Observations and modeling of seismic background noise. U. S. Geological Survey, Albuquerque, New Mexico. https://doi.org/10.3133/ofr93322.

Samuel H, Drilleau M, Rivoldini A, Xu ZB, Huang QC, Garcia RF, Lekić V, Irving JCE, Badro J, Lognonné PH, Connolly JAD, Kawamura T, Gudkova T and Banerdt WB (2023). Geophysical evidence for an enriched molten silicate layer above Mars's core. Nature **622**(7984): 712–717. https://doi.org/10.1038/s41586-023-06601-8.

Scholz JR, Widmer‐Schnidrig R, Davis P, Lognonné P, Pinot B, Garcia RF, Hurst K, Pou L, Nimmo F, Barkaoui S, De raucourt S, Knapmeyer‐endrun B, Knapmeyer M, Orhand‐mainsant G, Compaire N, Cuvier A, Beucler É, Bonnin M, Joshi R, Sainton G, Stutzmann E, Schimmel M, Horleston A, Böse M, Ceylan S, Clinton J, Van driel M, Kawamura T, Khan A, Stähler SC, Giardini D, Charalambous C, Stott AE, Pike WT, Christensen UR and Banerdt WB (2020). Detection, analysis and removal of glitches from InSight's seismic data from Mars. Earth Space Sci 7(11): e2020EA001317. https://doi.org/10.1029/2020EA001317.

Smith DE, Zuber MT, Neumann GA, Guinness EA and Slavney S (2003). Mars Global Surveyor Laser Altimeter mission

experiment gridded data record. NASA Planetary Data System, MGS-M-MOLA-5-MEGDR-L3-V1.0.

Stähler SC, Khan A, Banerdt WB, Lognonné P, Giardini D, Ceylan S, Drilleau M, Duran AC, Garcia RF, Huang QC, Kim D, Lekic V, Samuel H, Schimmel M, Schmerr N, Sollberger D, Stutzmann É, Xu ZB, Antonangeli D, Charalambous C, Davis PM, Irving JCE, Kawamura T, Knapmeyer M, Maguire R, Marusiak AG, Panning MP, Perrin C, Plesa AC, Rivoldini A, Schmelzbach C, Zenhäusern G, Beucler É, Clinton J, Dahmen N, Van Driel M, Gudkova T, Horleston A, Pike WT, Plasman M and Smrekar SE (2021). Seismic detection of the Martian core. Science **373**(6553): 443–448. https://doi.org/10.1126/science.abi7730.

Sun WJ, Wang YB, Wei Y and Zhao L (2021). Martian seismology and review of Martian interior structure. Rev Geophys Planet Phys **52**(4): 437–449. https://doi.org/10.19975/j.dqyxx.2021-016 (in Chinese with English abstract).

Wan WX, Wei Y, Guo ZT, Xu YG and Pan YX (2019). Toward a power of planetary science from a gaint of deep space exploration. Bull Chin Acad Sci **34**(7): 748–755. https://doi.org/10.16418/j.issn.1000-3045.2019.07.003 (in Chinese with English abstract).

Wang X, Chen L and Wang X (2023). Renewed epicentral distribution of low frequency marsquakes by varying-parameter polarization analysis of InSight data. Geophys Res Lett **50**(16): e2023GL103896. https://doi.org/10.1029/2023GL103896.

Xiao WB and Wang YB (2021). Discoveries of the InSight seismic investigation on Mars' surface. Rev Geophys Planet Phys **52**(2): 211–226. https://doi.org/10.19975/j.dqyxx.2020-022 (in Chinese with English abstract).

Xu WC, Zhu QW and Zhao L (2022). GlitchNet: A glitch detection and removal system for SEIS records based on deep learning. Seismol Res Lett **93**(5): 2804–2817. https://doi.org/10.1785/0220210361.

Zhang RQ, Geng Y, Sun ZZ, Li D, Zhong WA, Li HT, Cui XF and Liu JJ (2022). Technical innovations of the Tianwen-1 mission. Acta Aeron Astron Sin **43**(3): 626689. https://doi.org/10.7527/S1000-6893.2021.26689 (in Chinese with English abstract).